Sibling Influence on Study Choice

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Abstract

This paper studies spillovers from older to younger siblings' study choices. I use Dutch admission lotteries in medical schools and find that older siblings significantly influence their younger siblings' field of study choice. I show that 70% of siblings' common study choices are due to sibling spillovers, and the remaining 30% are due to common preferences. The family background rather than sibling personal characteristics appears to be the defining feature in explaining heterogeneous spillover effects.

Keywords: instrumental variable, siblings, study choice, the Netherlands

JEL Codes: I23, I24

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1 Introduction

How do people make decisions in their early life that will affect their adult outcomes? Who helps them to make these decisions? A field of study choice is one of such pivotal decisions that a person makes in life. This field of study choice is consequential in determining what career path to pursue. Many papers find a wide range of returns to different studies (Altonji et al., 2012, 2016). Usually, more math-oriented fields of study tend to produce higher returns than social sciences and humanities (Altonji et al., 2012). Further, certain fields, e.g. medicine and STEM, often have an insufficient number of students, though are seen as socially beneficial. Hence, a decision on what field to specialize in is crucial in determining labor market outcomes and affecting societal benefits. Less is known about how a person chooses a field of study and who can influence a person's study choice. Family networks and peers are found to influence important life choice decisions. Arguably, the closest peers to a person at a young age are one's siblings. Siblings can directly influence each other's behavior by simply observing what decisions they make in everyday life. Understanding how siblings can influence each other's educational choices is a relevant empirical question.

This paper studies how older siblings influence the study choice decisions of their younger siblings. Identifying peer effects is a challenging problem due to correlated effects and the reflection problem (Manski, 1993). I overcome these problems by exploiting a unique setting where admission to medical schools is determined by a lottery. I use administrative data on a universe of students who have ever participated in admission lotteries. My identification strategy relies on the fact that admission to medical schools is random, hence, any differences between the study choice of younger siblings, whose older siblings are lottery winners or lottery losers can be attributed to the influence of the older sibling's lottery outcome.

I find that older siblings have a significant and economically meaningful influence on younger siblings' field of study choice. I find that if older siblings win a lottery to a medical school, the probability of younger siblings being enrolled in a medical school increases by 3.2 percentage points. Usually, younger siblings choose a medical school 3.2% of the time, hence, this effect doubles the share of younger siblings in a medical school relative to the mean. A younger sibling is only 1.2 percentage

points more likely to be enrolled in the same program as an older sibling. Since younger siblings study in the same program as the older siblings 5.6% of the time, it translates to a 21.6% increase relative to the mean.

A simple model exploiting admission lotteries predicts competing effects of an older sibling's study choice on a younger sibling's study choice. In this model, there are two types of programs: lottery programs and programs without a lottery. Intuitively, if older siblings have a causal effect on their younger sibling's study choice, younger sibling would follow their older siblings regardless of the lottery outcome. In other words, younger siblings would follow their winning older siblings as likely as their losing older siblings. However, if common study choices of siblings are fully explained by their common educational preferences, younger siblings would choose a program that their older siblings initially applied. Even if older siblings are not accepted in the program to which they applied, the fact that they applied indicates a preference to study in this program. Since siblings have common preferences, younger siblings would also choose this program. Note that this choice is not attributed to sibling spillovers but explained by common preferences.

The highly stylized model makes key comparative static predictions very clear. Under the pure common preference effects, an older sibling's lottery outcome has a zero effect on whether a younger sibling is enrolled in a medical school and a positive effect on whether a younger sibling is enrolled in the same program as an older sibling. Under the pure sibling spillover effects, a model makes opposite predictions: a positive effect of older sibling's lottery outcome on the probability of younger sibling's enrollment in a medical school and a zero effect of older sibling's lottery outcome on the probability of younger sibling's enrollment to the same program as an older sibling. The results of this study provide evidence that older siblings have a direct causal effect on their younger siblings, while siblings' study preferences are correlated only to a minor extent within a family.

This paper relates to a limited but growing amount of papers studying siblings spillovers in universities (Altmejd et al., 2021; Aguirrea and Matta, 2021; Goodman et al., 2015) and secondary schools (Dahl et al., 2021; de Gendre, 2021; Gurantz et al., 2020; Dustan, 2018; Joensen and Nielsen, 2018). The closest in spirit to this study is a paper by Altmejd et al. (2021). They study sibling spillover effects in college and major choice across four countries. The authors use admission thresholds

in the RDD design to compare younger siblings' study choices whose older siblings were just above or below these cutoffs. They find that younger siblings follow their older siblings to the same college and the same college–major. Similarly, Aguirrea and Matta (2021) exploit discontinuous admission cutoffs in the RDD design generated by Chile's centralized higher education system and find strong sibling spillover effects on university enrollment but not on the field of study choice.

The paper has two main contributions to the literature. First, by exploiting the IV rather than RDD design, this study estimates sibling spillover effects for a wider population. To estimate sibling spillovers, previous papers exploit admission cutoffs that generate quasi–random variation in enrollment probabilities. Hence, they identify effects among older siblings who are close to the admission margin, i.e. marginal applicants. To identify causal effects for a larger and perhaps more representative group of individuals, one can use an IV design. In the paper, I use the unique institutional setting of admission lotteries in the IV setting which allows me to identify effects for inframarginal applicants.

Second, my findings shed light on the roots of differences in sibling spillovers. Previous papers exploited only the personal characteristics of siblings to explain the mechanisms of spillovers. I show that the family background is a driver of heterogeneous sibling spillover effects in the field of study choice. The findings of the paper suggest that studying together or role modeling mechanisms are not the driving force of sibling spillovers in study choices. If spillover effects are driven by the desire to study together, there should be stronger effects among closely spaced siblings. Yet, I find that closely spaced siblings are followed as likely as siblings who are further apart in age. Alternatively, younger siblings can follow older siblings since they serve as role models. Then one would expect that the same–sex siblings would resemble more similar study choices since the field of study–specific information might be more relevant among them. Additionally, older siblings with higher abilities can be argued to be followed more than siblings with lower abilities, since they serve as academically better role models. Despite this, same-sex siblings or higher–ability older siblings are not followed more often. I show that parental investment in children explains the differences in spillover effects. Older siblings in poorer families have less influence on younger siblings' study choices. Notably, siblings spillover effects are not existent in single–parent families. Taken together,

these findings suggest that due to less parental investment of income and time in children, the quantity of sibling interactions decreases due to lower quality of sibling's education advice.

Understanding the nature of sibling spillover matters for educational reforms. The common preference effects and the sibling spillover effects have two different policy implications. If sibling common study choices are due to common preferences, a policy maker cannot directly influence the decision of choosing a specific field. Common preferences are formed before the application process, which makes it hard to change one's decisions. If the common study choices are due to sibling spillovers, programs and campaigns aimed at increasing enrollment in specific fields of study would have multiplicative effects. From a policy perspective, the results of the paper imply that additional information about the program can help in attracting more students into these fields. However, the policy recommendations should be applied with caution. Older siblings from poorer families and single–parent families show no influence on younger siblings' study choices. This raises a concern about whether strong family networks provide effective education support for the poor and single–parent families.

The paper is structured as follows. Section 2 presents a brief description of the Dutch higher education system and lottery system. Section 3 presents a stylized model of study choice and derived testable predictions. Section 4 presents data. Section 5 shows my identification strategy. Section 6 presents the main results, possible mechanisms of sibling spillovers, and findings of the simulation analysis. Section 7 presents concluding remarks and discussion.

2 Institutional background

In the Netherlands, students who complete the academic track in secondary school can enroll in any Dutch university. The majority of university programs have to accept all students, while some programs have quotas with a limited number of places, e.g. medical schools.

Until 1999, medical schools used a nationwide centralized lottery to admit students. The lottery was introduced as a response to the increasing number of applicants to medical schools which exceeded the number of available places. The

Ministry of Education determines quotas. If the number of applicants exceeds the number of available places, a lottery determines who is admitted. The lottery is weighted, so students with a higher GPA have a higher probability of winning. A GPA is determined based on the grades from nationwide and externally graded high school exams, where six indicates a pass. Table 1 shows the weights and shares of groups of students in different lottery categories. For a more detailed description of Dutch admission lotteries and the educational system, see Ketel et al. (2016).

Table 1: Lottery categories

Category	GPA	Share	Weight
A	$GPA \geq 8.5$	0.02	2.00
B	$8.0 \leq GPA < 8.5$	0.05	1.50
C	$7.5 \leq GPA < 8.0$	0.09	1.25
D	$7.0 \leq GPA < 7.5$	0.21	1.00
E	$6.5 \leq GPA < 7.0$	0.22	0.80
F	$GPA < 6.5$	0.30	0.67
Other	–	0.11	1.00

Notes: A GPA is a grade point average on the final exams in high school. Share is the share of applicants in different categories that applied for the lotteries in the years 1988-1999. Weight indicates the relative probability of being admitted. The category "other" refers to students who did not participate in the nationwide high school exams, such as foreign students. This category is excluded from the analysis.

3 Predictions

There are multiple reasons why siblings follow similar educational pathways. First, siblings can have common educational preferences. Siblings are exposed to a similar environment, e.g. family, and they might attend the same school with similar teachers and classmates. As a result of such a close connection, it is quite common for siblings to choose a similar field of study not because siblings have a causal effect on each other, but due to a similar environment. This is called the common preference effect.

Second, an older sibling can have a direct influence on a younger sibling. Older siblings can provide valuable information about a field she is enrolled in. For exam-

ple, an older sibling can inform a younger sibling about specific courses, the quality of peers, and the level of a program which can be helpful for a younger sibling to choose a field. This type of information is hard to observe without studying in a program. Older siblings lower the costs of obtaining such information and make study options more salient to their younger siblings. Alternatively, older siblings can serve as role models, and younger siblings would simply imitate them regardless of the information provided. This is called the sibling spillover effect.

To distinguish between common preference effects and sibling spillover effects, I use admission lotteries to medical schools. Admission lotteries allow me to estimate two treatment effects:

1. The effect of an older sibling's lottery outcome on whether a younger sibling is enrolled in the program where an older sibling has *applied*. Since medical schools are the only lottery programs in this study, I call this an effect of the older sibling's lottery outcome on whether the younger sibling is enrolled in *medical school*.

$$\Delta_{MS} = \mathbb{E}[MedicalSchool_{YS}|Winner_{OS}] - \mathbb{E}[MedicalSchool_{YS}|Loser_{OS}]$$

2. The effect of an older sibling's enrollment in a lottery program on whether a younger sibling is enrolled in the program where an older sibling is *enrolled*. I call this effect an effect of the older sibling's lottery outcome on whether the younger sibling is enrolled in the *same program*.

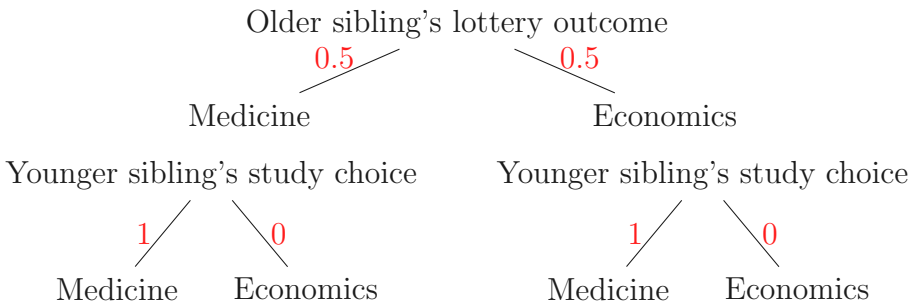
$$\Delta_{SP} = \mathbb{E}[SameProgram_{YS}|Winner_{OS}] - \mathbb{E}[SameProgram_{YS}|Loser_{OS}]$$

Let us analyze a stylized example of a study choice model to predict these two treatment effects. Assume there are two study programs: medicine and economics. Medicine is a program that uses a lottery to admit students. There is a 50% probability of winning a lottery. Assume all lottery winners comply with the outcome of a lottery, i.e. full compliance. Economics is a program without a lottery, i.e. all students who apply are accepted.

First, let us consider a case under the common preference effects in Figure 1. When siblings have common educational preferences. Since all older siblings in the sample applied to a lottery, they revealed preferred medicine as their study choice.

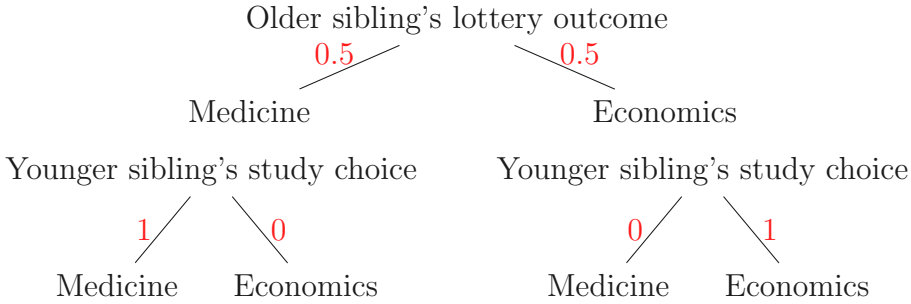
Under the common preference effect, the study preferences of younger siblings are identical among older siblings who are lottery winners and lottery losers. Hence, when the common preference effect is prevailing, the effect of an older sibling's lottery outcome on whether a younger sibling is enrolled in a medical school is predicted to be *zero*, while the effect on enrollment in the same program is predicted to be *positive*.

Figure 1: A model of study choice under common preference effects



Second, let us consider a case under the sibling spillover effect in Figure 2. When older siblings provide information about a study program or serve as role models. As a result, younger siblings are enrolling in the programs of their older siblings regardless of their lottery outcome. Hence, under the sibling spillover effect, the effect of an older sibling's study choice on whether a younger sibling is enrolled in the medical school is predicted to be *positive*, while the effect on enrollment in the same program is predicted to be *zero*.

Figure 2: A model of study choice under sibling spillover effects



The summary of model predictions is presented in Table 2. Under common preferences and sibling spillover models, the magnitude of treatment effects differs,

which allows me to empirically test which effect is prevailing. These predictions are the main contribution to the literature as I can test directly to what extent sibling common study choices are explained by their common preferences or by the causal effect of older siblings.

Table 2: Predicted treatment effects of an older sibling's lottery outcome on a younger sibling's study choice

Outcome	Common preferences	Sibling spillovers
Δ_{MS}	0	+
Δ_{SP}	+	0

4 Data

4.1 Data source and sample

I use administrative data from the Statistics Netherlands (CBS) which allow linking all individuals to their parents and siblings. The registry contains information on different demographic and educational outcomes, i.e. age, sex, immigration background, university enrollment, field of study, and lottery outcomes. I observe lottery data on individuals who applied to medical schools between 1987 and 2004. For lottery participants, I observe the year when they participated in a lottery, lottery category, and lottery outcome.

To ensure that I observe first-year applicants, I exclude applicants who participated in the lottery in 1987 since there is no information about lottery participation in previous years. Since the lottery system was gradually abandoned from 1999 onward, I exclude lottery participants applying for the first time after this year. Applicants from category A are also excluded since almost all of them are admitted to a medical school. Further, I exclude applicants older than 20 at the moment they first participated in a lottery to exclude individuals who did not follow the usual educational track. Finally, I exclude applicants admitted via ministers' places since they do not participate in a lottery.

A unit of the analysis is a sibling pair. Hence, for every lottery participant, I link their younger siblings using a registry of their parents. I restrict the sample to lottery applicants who have at least one younger (half)–sibling. I include siblings who are born at least one year and not more than twenty years apart. In case there are several older siblings in a family participating in a lottery, I take the oldest older sibling to determine the treatment status of a younger sibling. The results are identical when I use the youngest older sibling, when I include families with only one older sibling participating in a lottery, or when I restrict the sample only to two sibling families.

4.2 Descriptive statistics

Table 3 shows the balancing table of pre–determined characteristics of older and younger siblings by the outcome of the first lottery of older siblings. The table presents sample means, and p–values obtained from regressing personal characteristics on an indicator function for an older sibling winning the first lottery, including fixed effects for the program, lottery category, year of the first lottery participation, and their interactions. Around 60% of older siblings are female, 7% have a non–western background, and the average age at the first lottery is 18.8. The average number of siblings in a family is 3. Personal characteristics of older siblings are well balanced as indicated by p–values.

Table 3 also represents the personal characteristics of younger siblings. Around 50% of the younger siblings are female, 8% have a non-western background, and the average age difference between younger and older siblings is 4.5 years. Around 50% of sibling pairs are same–sex pairs.

Table 4 presents descriptive statistics of enrollment of older and younger siblings. Almost all older siblings are enrolled in a university: 99% among winners and 95% among losers. Winning a lottery is associated with a 50 percentage point increase in enrollment in a lottery program for older siblings. For younger siblings, around 82% are enrolled in a university, while 9% are enrolled in a lottery program.

Table 3: Balancing of personal characteristics of older and younger siblings by the outcome of the first lottery of older siblings

Outcome	Lottery winners	Lottery losers	p-value
Older siblings			
Female	0.59	0.59	0.62
Non-western immigrant	0.07	0.07	0.39
Age at first application	18.82	18.86	0.74
Family size	2.97	2.97	0.42
N	8029	10002	
Younger siblings			
Female	0.49	0.48	0.02
Non-western immigrant	0.08	0.09	0.13
Age difference in years	4.50	4.43	0.35
Same sex pairs	0.49	0.50	0.21
N	12294	15348	

Note: The p–values in the final column are from the regressing personal characteristics on an indicator function for an older sibling winning the first lottery, including fixed effects for program, lottery category, year of the first lottery participation, and their interactions.

5 Empirical approach

I estimate the effect of an older sibling's lottery outcome on a younger sibling's study choice, using the following model:

$$Y_i = \delta D_i + X_i\beta + \alpha_i + U_i \tag{1}$$

where Y_i indicates if in a sibling-pair i a younger sibling is enrolled in the 1) medical program and 2) same program as an older sibling; D_i indicates the result of the first lottery of an older sibling; X_i includes indicators for siblings' sex and non-western origin, age of an older sibling at the first lottery; α_i are the program, lottery category, year of the first lottery participation fixed effects, and their interactions.

The effect of an older sibling's lottery outcome on a younger sibling's study choice is captured by δ. δ is the reduced–form estimate which captures the increased probability of a younger sibling enrolling in the program if an older sibling wins a lottery. Since the program admission of an older sibling is random conditional on the program, lottery category, year of the first lottery participation fixed effects, and

Table 4: Enrollment outcomes of older and younger siblings by the outcome of the first lottery of older siblings

Outcome	Lottery winners	Lottery losers
Older siblings		
Enrolled in a university	0.99	0.95
Enrolled in a lottery program	0.95	0.44
N	8029	10002
Younger siblings		
Enrolled in a university	0.83	0.81
Enrolled in a lottery program	0.10	0.08
Enrolled in the same university	0.22	0.21
Enrolled in the same program	0.07	0.06
N	12294	15348

Note: Since the lottery is weighted, the observed differences between lottery winners and losers in this table cannot be interpreted as causal.

their interactions, the zero mean conditional independence assumption holds for the first lottery: $\mathbb{E}[U_i|X_i, \alpha_i, D_i] = \mathbb{E}[U_i|X_i, \alpha_i]$.

Admission lotteries help me to overcome the reflection problem as the variation in the older sibling's study choice is exogenously determined by a lottery and is not influenced by the younger sibling's study choice. I cluster standard errors at the family level, as treatment is assigned at this level and there can be multiple siblings within a family.

6 Results

6.1 Sibling spillovers on the field of study choice

This section presents the main results of whether older siblings influence younger siblings' field of study choice.

Table 5 shows reduced form estimates of sibling spillovers on the field of study choice. If an older sibling wins a lottery, a younger sibling's probability to be enrolled in a medical school increases by 3.2 percentage points. This doubles the share of siblings enrolled in a medical school compared to the control mean of 3.2% (which is reported in parentheses throughout the rest of the paper). A younger sibling is

1.2 percentage points more likely to be enrolled in the same program as an older sibling (5.6%). Based on the predictions derived earlier, these results indicate that the sibling spillover effect is stronger than the common preference effect.

In other words, younger siblings, whose older siblings lost a lottery, are less likely to enroll in a medical school and more likely to follow their older sibling's study program choice. This indicates that common educational preferences formed before the realization of the lottery outcome are not strong in explaining sibling common study choices. These findings indicate that older siblings have a direct causal effect on their younger siblings.

Table 5: Reduced–form estimates of the effect of older sibling's lottery outcome on younger sibling's study choice

Outcome	Younger sibling is enrolled in the	
	Medical school	Same program
Older sibling wins a lottery	0.032***	0.012***
	(0.003)	(0.004)
Control mean	0.032	0.056
Effect size	98.7%	21.6%
N	27642	27642

Note: Standard errors are clustered at the family level. All regressions control for siblings' sex, non-western origin, age of an older sibling at the first lottery, fixed effects for program, lottery category, year of the first lottery participation, and their interactions. *p<0.1; **p<0.05; ***p<0.01

6.2 Mechanisms

This section presents suggestive evidence of mechanisms explaining sibling spillovers, i.e. studying together, role modeling, parental investment, and cultural values.

Studying together

If siblings assign some value to study together with their older siblings, spillover effects should disappear if siblings are far apart in terms of age. To test this, Table 6 presents the results of sibling spillovers for siblings who are more than six years apart

from each other, hence, they are unlikely to study together. The estimates show that sibling spillovers are not affected by the age distance between siblings. The fact that spillover effects do not disappear if there is more than 6 years difference between siblings indicates that sibling spillovers are not driven by the desire of younger siblings to study together with older siblings.

Role modeling

Older siblings can serve as role models to their younger siblings. It can be argued that the experience of siblings of the same sex is more relevant in choosing a field of study. If siblings serve as role models, there should also be stronger spillover effects from higher ability older siblings as they serve as a better example of academically successful students. Hence, younger siblings would be better off following their more successful older siblings than less successful ones. To explore role modeling as a possible channel, I look at the heterogeneity of sibling spillovers between same–sex and opposite–sex siblings as well as older siblings with higher and lower GPAs.

Table 7 presents the results of sibling spillovers by the sex composition of sibling pairs. Overall, there is not much systematic heterogeneity of sibling spillovers among siblings of the same and different sexes. Table 8 shows the results by the ability of older siblings. As a proxy of ability, I use the lottery category of an older sibling. Note, the higher the high school GPA of an older sibling, the higher the lottery category. I interact the lottery outcome of an older sibling with an indicator for having a GPA higher than 7.5. Sibling spillovers are similar across older siblings of different ability levels. These findings provide suggestive evidence against siblings serving as role models.

Parental investment

Parental investment in children depends on available parental resources, i.e. income and time. Poorer families have a lower income to invest in their children. Single parents have less time to devote to their children due to more stringent time constraints. Fewer available parental resources translate into less investment in children. Hence, the quality of children decrease in families with less available resources (Becker and Tomes, 1976). The reduction in the quality of children decreases the

benefits of sibling interactions. As a result, social interactions of siblings, such as asking for advice about education trajectories, are less likely to occur among families with fewer available resources.

To explore whether sibling spillovers are driven by the differences in parental investment in children, Table 9 investigates differences in sibling spillovers by parental income and Table 10 shows sibling spillovers in single– and two–parent families. Sibling spillovers are significantly lower among poorer families. Siblings' common preferences are also correlated to a lesser extent in poorer families. A stronger pattern is observed for single–parent families. Older siblings do not influence younger siblings' study choices in single–parent families nor their education preferences are correlated. These findings indicate that due to lower parental investment of income and time, sibling spillovers decrease.

Cultural values

Table 11 presents heterogeneous results by the migration background. Overall, siblings spillovers are stronger among siblings with a non–western migration background. These findings suggest that migrant families rely more heavily on the experience of older siblings in study choice decisions due to a possible lack of information. Interestingly, sibling preferences are also correlated to a higher extent among families with a migrant background. One explanation is that family members with migrant backgrounds depend more heavily on their families in general. This might potentially be detrimental to children's education if education inequality persists across generations.

In sum, sibling spillovers in the field of study choices do not differ by the age difference, sex composition, and the ability of older siblings. Spillover effects do vary by parental income, single–parent families, and migration status. Ultimately, the analysis of heterogeneous effects is consistent with siblings proving valuable advice about study choices. Older siblings from better–endowed families provide higher quality advice about study trajectories. I rule out the studying together and role modeling channels as a possible explanation of sibling spillovers in study choice.

6.3 Robustness checks

Table 12 presents the results of sibling spillover effects using different sample definitions. In the main analysis, I use the oldest older sibling to determine the treatment status of all younger siblings, and these are baseline results in Column (1). In Column (2), I use the younger oldest sibling, in Column (3) – only families with one older sibling participating in a lottery, in Column (4) – only families with two siblings, and in Column (5), I do not put an upper bound on the age difference between siblings. Overall, the results are robust and stay virtually unchanged using different sample definitions.

6.4 Simulation

The findings of the paper show that sibling spillover effects exist. This section provides results from the simulation analysis to determine the relative role of common preference effects and sibling spillover effects.

Model

I follow the analysis of the stylized model described in Section 2. Note that the common preferences and sibling spillover effects lead to the same field of study choice of younger siblings whose older siblings are lottery winners: younger siblings always choose a medical school which is the program their older siblings are enrolled in. Since I am interested in disentangling the relative role of common preferences and sibling spillovers, I focus on simulating the study choice of younger siblings whose older siblings are lottery losers.

As before, younger siblings follow where their older siblings have applied or are enrolled. Under the pure common preference effects, younger siblings choose a program where their older siblings have applied, i.e. a medical program. Under the pure sibling spillover effects, younger siblings choose a program where their older siblings are enrolled, i.e. the same program. Hence, the higher the role of sibling spillovers, the more likely younger siblings would choose the program their older siblings are enrolled in.

To simulate these fields of study choice, let me introduce some notations. Let

Ω_{OS} denote a set of older sibling's field of study programs. Let Ω_{YS} denote a set of younger sibling's field of study programs. Let p_i be the probability that younger siblings would choose a program from the set of older sibling's study programs Ω_{OS}. Then with the probability $p_i \in [0, 0.05, ..., 0.95, 1]$ a younger sibling i chooses a program from a set Ω_{OS} and with the probability $1 - p_i$ – from a set Ω_{YS}. In other words, p_i is the parameter determining the relative strength of sibling spillovers, i.e. *sibling influence*. For example, when $p_i = 0$, the younger sibling's study choice is fully determined by common preferences. When $p_i = 1$, the younger sibling's study choice is fully determined by sibling spillovers.

To determine the role of the sibling influence in determining common study choices:

1. I vary p_i between 0 and 1.

2. For each value of p_i, I draw study choices of younger siblings from actual study choices, following the procedure described above.

3. I estimate Model 1 for each draw to get reduced–form estimates of the older sibling's lottery outcome on the younger sibling's study choice.

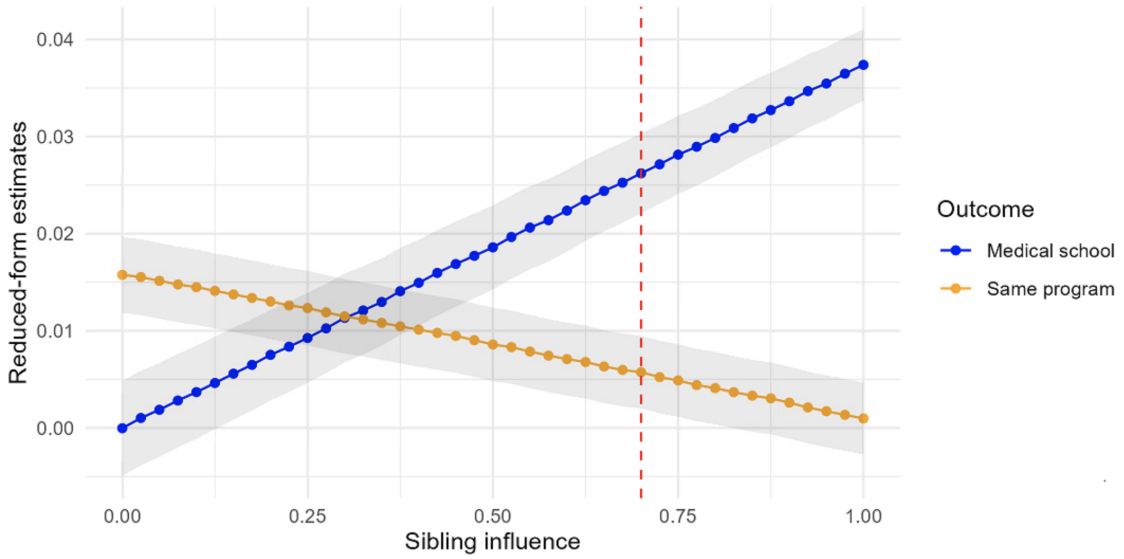
4. I repeat this process 1000 times for each p_i and take an average of point estimates and 95% confidence interval.

Results

The results of the simulation analysis are in Figure 3. The blue line shows reduced–form estimates of the effect of an older sibling's lottery outcome on a younger sibling's enrollment in a medical school. The yellow line shows reduced–form estimates of the effect of older siblings' lottery outcome on younger siblings' enrollment in the same program as their older siblings. The colored area is a 95% confidence interval. The red line represents the location of reduced–form estimates found in the main analysis.

As predicted by the stylized model, when the common preference effect is prevailing, i.e. $p_i = 0$, the effect of older sibling lottery outcome on whether a younger sibling is enrolled in a medical school is zero, i.e. $\Delta_{MS} = 0.000$ (s.e. $= 0.002$), while

Figure 3: Reduced–form estimates of the effect of older sibling's lottery outcome of younger sibling's enrollment from 1000 simulations



Note: A red line shows the location of reduced–form estimates of sibling spillovers presented in the main analysis.

the effect on whether a younger sibling is enrolled in same the program as an older sibling is positive, i.e. $\Delta_{SP} = 0.016$ (s.e. $= 0.002$). On the other hand, when the sibling spillover effect is prevailing, i.e. $p_i = 1$, the effect of an older sibling's lottery outcome on whether a younger sibling is enrolled in a medical school is positive, i.e. $\Delta_{MS} = 0.037$ (s.e. $= 0.002$), while the effect on enrollment in the same program as an older sibling is zero, i.e. $\Delta_{SP} = 0.000$ (s.e. $= 0.002$).

The reduced–form estimates of sibling spillovers found in the main analysis in Table 5 are as follows: $\Delta_{LP} = 0.032$ (s.e. $= 0.003$) and $\Delta_{SP} = 0.012$ (s.e. $= 0.004$). Comparing these estimates to the results of the simulation analysis, I find that 70% of sibling common study choices are due to sibling spillovers and the remaining 30% are due to common preferences.

7 Conclusion and discussion

The results of the paper provide evidence of strong sibling spillover effects in study choice. These findings have relevant policy implications. Policymakers can use information campaigns to better inform students of their study options, which would lead to multiplicative effects due to spillovers. However, if one aims at targeting disadvantaged students, i.e. students from poorer families or students from single–parent families, such programs can be less effective in achieving desired goals.

Interestingly, previous papers find no sibling spillovers on the field of study choice (Altmejd et al., 2021; Aguirrea and Matta, 2021). In contrast, I find substantial positive effects on the field of study choice. There are two possible explanations for these differences. First, previous papers studying sibling spillover effects use RDD design exploiting admission cutoffs. The estimated effects are for younger siblings whose older siblings are very close to the cutoff points, i.e. marginally admitted students. This study uses an IV design exploiting admission lotteries. The estimated effects are for younger siblings whose older siblings won the lottery, i.e. inframarginal students. Perhaps, younger siblings gain little from marginally accepted older siblings in terms of field of study choice, while they benefit a lot from inframarginal older siblings. Hence, one should be cautious in generalizing the existence and nature of sibling spillover effects in different settings.

Second, note that the average enrollment in universities in my sample is 80–90%, while in previous studies it is ranging from 15–18% in Chile and Croatia to 35–40% in Sweden and the United States. The pattern of these results suggests sibling spillovers might depend non–trivially on institutional characteristics. Speculatively, in settings with high enrollment in universities sibling spillovers might be less important for a decision to enroll in a university (extensive margin decisions) and more important for what to study (intensive margin decisions). Hence, further research is needed to make conclusions about how sibling spillovers depend on the features of higher education systems.

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Appendix

Table 6: Reduced–form estimates of the effect of older sibling's lottery outcome on younger sibling's study choice by the age difference

Outcome	Younger sibling is enrolled in the	
	Medical school	Same program
Older sibling wins a lottery		
	()	()
Older sibling wins a lottery × Age difference > 6	()	()
N		

Note: Standard errors are clustered at the family level. All regressions control for siblings' sex, non-western origin, age of an older sibling at the first lottery, fixed effects for program, lottery category, year of the first lottery participation, and their interactions. In addition they include interactions with an indicator for whether siblings are six of more years apart. *p<0.1; **p<0.05; ***p<0.01

Table 7: Reduced–form estimates of the effect of older sibling's lottery outcome on younger sibling's study choice by the sex composition

Outcome	Younger sibling is enrolled in the	
	Medical school	Same program
Older sibling wins a lottery		
	()	()
Older sibling wins a lottery × Same sex	()	()
N		

Note: Standard errors are clustered at the family level. All regressions control for non-western origin, age of an older sibling at the first lottery, fixed effects for program, lottery category, year of the first lottery participation, and their interactions. In addition they include interactions with an indicator for whether siblings are of the same sex. *p<0.1; **p<0.05; ***p<0.01

Table 8: Reduced–form estimates of the effect of older sibling's lottery outcome on younger sibling's study choice by ability of older siblings

Outcome	Younger sibling is enrolled in the	
	Medical school	Same program
Older sibling wins a lottery		
	()	()
Older sibling wins a lottery × GPA > 7.5	()	()
N		

Note: Standard errors are clustered at the family level. All regressions control for siblings' sex, non-western origin, age of an older sibling at the first lottery, fixed effects for program, lottery category, year of the first lottery participation, and their interactions. In addition they include interactions with an indicator for whether older sibling's lottery category is B or C, i.e. GPA is higher than 7.5. *p<0.1; **p<0.05; ***p<0.01

Table 9: Reduced–form estimates of the effect of older sibling's lottery outcome on younger sibling's study choice by parental income

Outcome	Younger sibling is enrolled in the	
	Medical school	Same program
Older sibling wins a lottery		
	()	()
Older sibling wins a lottery × Parental income < median	()	()
N		

Note: Standard errors are clustered at the family level. All regressions control for siblings' sex, non-western origin, age of an older sibling at the first lottery, fixed effects for program, lottery category, year of the first lottery participation, and their interactions. In addition they include interactions with an indicator for whether parental income is lower than the median parental income. *p<0.1; **p<0.05; ***p<0.01

Table 10: Reduced–form estimates of the effect of older sibling's lottery outcome on younger sibling's study choice by single and two parents families

Outcome	Younger sibling is enrolled in the	
	Medical school	Same program
Older sibling wins a lottery		
	()	()
Older sibling wins a lottery × Single–parent family	()	()
N		

Note: Standard errors are clustered at the family level. All regressions control for siblings' sex, non-western origin, age of an older sibling at the first lottery, fixed effects for program, lottery category, year of the first lottery participation, and their interactions. In addition they include interactions with an indicator for whether siblings live in single–parents families, when younger siblings are 14–18 years old. *p<0.1; **p<0.05; ***p<0.01

Table 11: Reduced–form estimates of the effect of older sibling's lottery outcome on younger sibling's study choice by migration background

Outcome	Younger sibling is enrolled in the	
	Medical school	Same program
Older sibling wins a lottery		
	()	()
Older sibling wins a lottery × Non–western background	()	()
N		

Note: Standard errors are clustered at the family level. All regressions control for siblings' sex, age of an older sibling at the first lottery, fixed effects for program, lottery category, year of the first lottery participation, and their interactions. In addition they include interactions with an indicator for whether older siblings have non–western background. *p<0.1; **p<0.05; ***p<0.01

Table 12: Reduced–form estimates of the effect of older sibling's lottery outcome on younger sibling's enrollment with different sample definitions

	Outcome	Oldest OS	Youngest OS	One OS	Two siblings	No age limit
1	Medical school	0.032***	0.031***	0.029***	0.030***	0.031***
		(0.003)	(0.003)	(0.003)	(0.006)	(0.003)
	Control mean	0.032	0.032	0.031	0.026	0.032
	Effect size	98.7%	96.4%	94.4%	115.0%	98.4%
2	Same program	0.012***	0.012***	0.010***	0.010	0.012***
		(0.004)	(0.003)	(0.003)	(0.008)	(0.003)
	Control mean	0.056	0.056	0.055	0.059	0.055
	Effect size	21.6%	20.5%	18.3%	17.6%	22.2%
	N	27642	27642	26646	7195	28222

Note: Standard errors are clustered at the family level. All regressions control for siblings' sex, non-western origin, age of an older sibling at the first lottery, fixed effects for program, lottery category, year of the first lottery participation, and their interactions. *p<0.1; **p<0.05; ***p<0.01